

# MUHAMMAD MUSTAFA

Software Engineer | AI/ML Engineer | Full Stack Developer

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## PROFESSIONAL SUMMARY

Final-year Computer Science student with hands-on experience in Software Engineering, Artificial Intelligence, Machine Learning, and Full-Stack Web Development using React.js, Node.js, and Express.js. Proficient in RESTful API design, relational and non-relational databases (PostgreSQL, MongoDB, Oracle SQL), and Object-Oriented Programming. Proven leadership in national-level tech events. Seeking software engineering or AI/ML internship roles to deliver scalable, production-ready solutions.

## TECHNICAL SKILLS

**Programming Languages:** C, C++, Python, JavaScript, TypeScript, HTML, SQL

**Frontend Development:** React.js, Tailwind CSS, Responsive Web Design, Vite

**Backend Development:** Node.js, Express.js, RESTful API Development

**Databases:** PostgreSQL, MongoDB, Oracle SQL, Supabase

**Artificial Intelligence / Machine Learning:** Supervised Learning, A-Star Search, Uniform Cost Search, Dijkstra Algorithm, NetworkX, Scikit-learn

**Tools and Methodologies:** Git, GitHub, Object-Oriented Programming, Design Patterns, Agile, Streamlit

## EDUCATION

**Bachelor of Science in Computer Science - FAST-NUCES, Karachi**

Aug 2023 - Jun 2027 (In Progress) | CGPA: 3.05 / 4.0

## WORK EXPERIENCE

**Teaching Assistant (TA) at FAST-NUCES, Karachi**

Aug 2025 - May 2026

- Supported 100+ students in Linear Algebra, Calculus, and Database Systems, improving comprehension and academic performance.
- Provided targeted guidance on debugging, algorithm analysis, and problem-solving strategies using data structures.
- Evaluated assignments and laboratory work, delivering structured feedback to enhance student learning outcomes.

## TECHNICAL PROJECTS

**Algo Rank - Competitive Programming Platform**

*Technologies: React.js, Node.js, Express.js, PostgreSQL, RESTful API*

- Designed and developed a full-stack real-time coding competition platform with dynamic leaderboards, live scoring, and JWT-based authentication.
- Built scalable backend services supporting 200+ concurrent participants with secure problem submission and session management.

**Karachi Market Intelligent Path Finder - AI Navigation System**

*Technologies: Python, Streamlit, A-Star Search, Uniform Cost Search, API Integration*

- Developed an AI-based navigation system applying A-Star and Uniform Cost Search on real geographical data, achieving 40% faster computation.

**Graph-Based Shortest Path Analyzer - AI/ML**

*Technologies: Python, Machine Learning, Dijkstra Algorithm, NetworkX, Scikit-learn*

- Implemented optimized shortest-path computation on weighted graphs; trained supervised machine learning model achieving 85% accuracy in travel time prediction.

**Bank Management System**

*Technologies: C++, Object-Oriented Programming, Data Structures, Design Patterns*

- Built a full-featured banking application with account management, fund transfers, and loan processing using Factory and Singleton design patterns.

**Air Ticket Reservation System**

*Technologies: React.js, Node.js, Express.js, Oracle SQL*

- Developed a flight booking platform with real-time seat availability, payment integration, and a normalized Oracle SQL database schema.

## LEADERSHIP AND ACTIVITIES

**Co-Head - Extended Procom (Media Production) | FAST-NUCES | 2025-2026**

- Led 10+ member team producing multimedia content for Procom, Pakistan's largest computing event; oversaw design, animation, and video production.

**Deputy Organizer - AI Competitions (Vibe Coding) | Procom | 2026**

- Organized AI-focused coding competitions attracting 200+ nationwide participants; managed logistics, problem design, and judging criteria.

**Coordinator - ACM Animation Team | ACM FAST Chapter | 2024-2025**

- Produced and edited 15+ promotional videos for university events and workshops, consistently delivering under tight deadlines.

## ACHIEVEMENTS AND CERTIFICATIONS

- Star Performer Award - ACM FAST Chapter (Technical Contribution and Leadership)
- Shields Award - Procom FAST (Event Management and Team Coordination)
- Star Performer Award - Procom FAST (Leadership and Technical Excellence)
- Multiple Competition Certificates - Procom, DevDay, Coders Cup (2024-2026)